MINI PROJECT REPORT

SMS Spam Classifier Using Machine Learning

Subject	Machine Learning
Dataset	UCI SMS Spam Collection (5,572 messages in the downloaded file)
Techniques	TF-IDF feature extraction and Multinomial Naive Bayes
Environment	Python in Google Colab

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1. Abstract

This project builds a supervised machine learning model to classify SMS messages as spam or ham (legitimate messages). The UCI SMS Spam Collection dataset is loaded into Python using Pandas. Messages are split into training and testing sets in an 80:20 ratio, converted to numerical TF-IDF features, and used to train a Multinomial Naive Bayes classifier. The model is evaluated using accuracy, spam precision, spam recall, a classification report, and a confusion matrix. Finally, three new SMS examples are passed to the model for prediction.

2. Aim and Objectives

Aim: To train and evaluate a text classification model that identifies spam SMS messages.

Objectives

• Load and inspect a labelled SMS dataset.
• Examine the distribution of spam and ham messages.
• Split the data into 80% training and 20% testing sets.
• Convert message text into TF-IDF numerical features.
• Train a Multinomial Naive Bayes classifier.
• Evaluate performance using accuracy, precision, recall, and a confusion matrix.
• Test the model on three new SMS messages.

3. Tools and Libraries

Tool / library	Purpose
Google Colab	Cloud notebook environment for running Python.
Pandas	Loading and inspecting the dataset.
Scikit-learn	Train-test splitting, TF-IDF, Naive Bayes, and evaluation metrics.
Matplotlib	Visualizing class distribution and confusion matrix.
Joblib	Saving the trained model and TF-IDF vectorizer for reuse.

4. Dataset Description

The notebook loaded 5,572 SMS messages. The dataset has two columns: `label`, with values `ham` or `spam`, and `message`, containing the SMS text. The notebook output reports no missing values in either column.

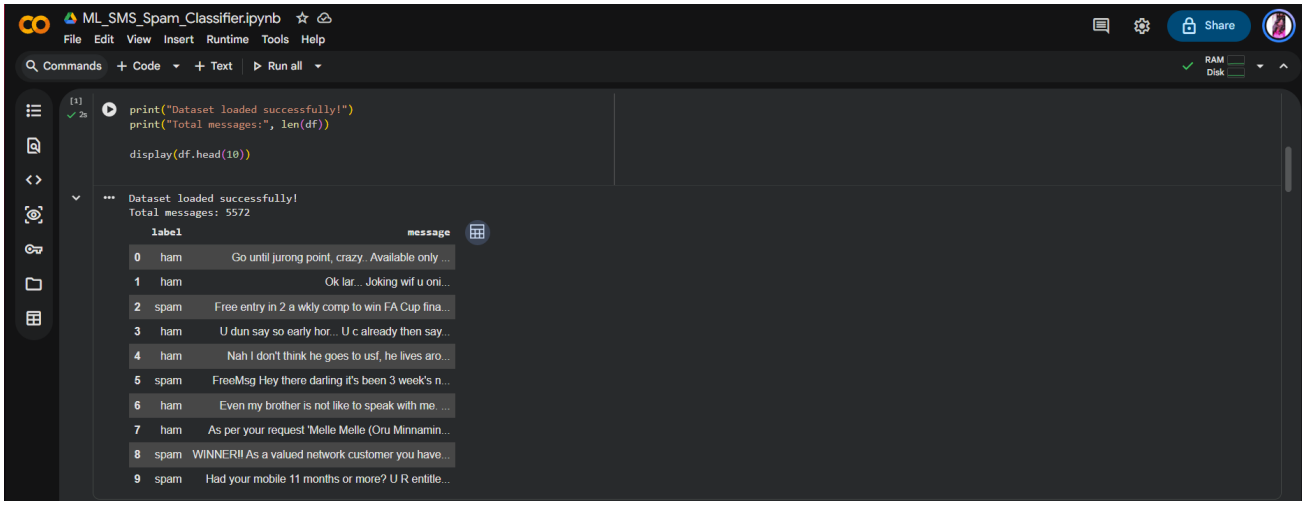


Figure 1. Dataset loaded in Colab, including a preview of the first ten messages.

5. Class Distribution

The displayed class counts are 4,825 ham messages and 747 spam messages. The dataset is imbalanced, with substantially more legitimate messages than spam messages. This is why precision and recall are useful alongside accuracy.

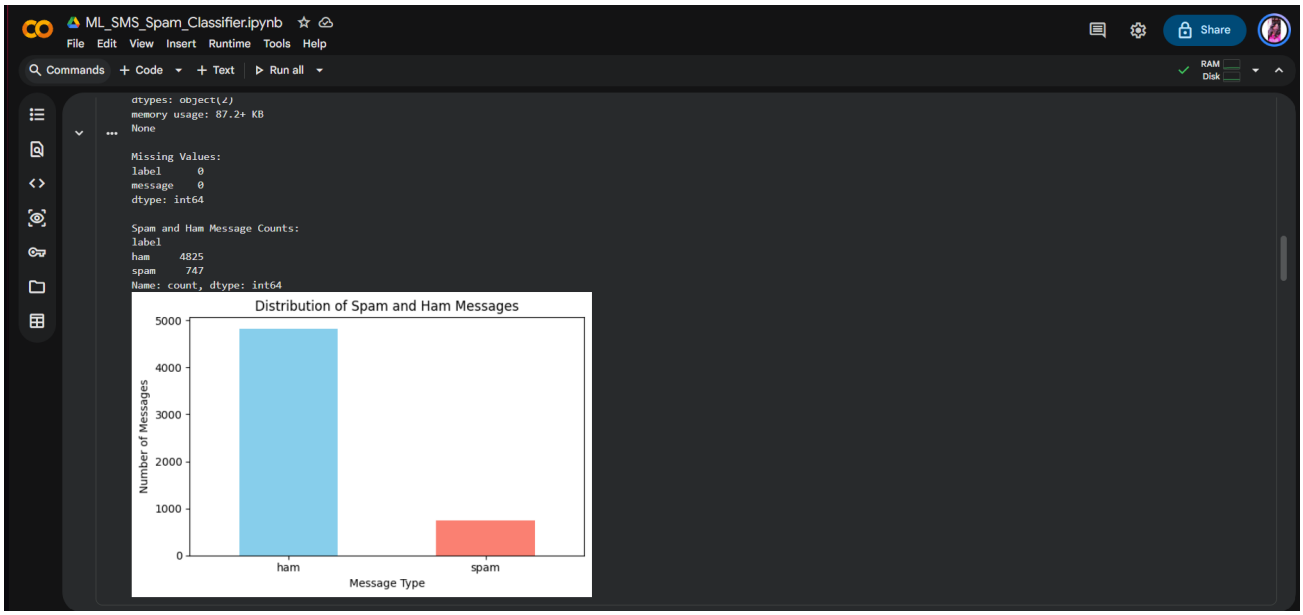


Figure 2. Distribution of ham and spam messages.

6. Train-Test Split

The dataset was split using `train_test_split` with `test_size=0.20`, `random_state=42`, and stratification by label. The output shows 4,457 training messages and 1,115 testing messages. Stratification preserves the approximate class balance across both subsets.

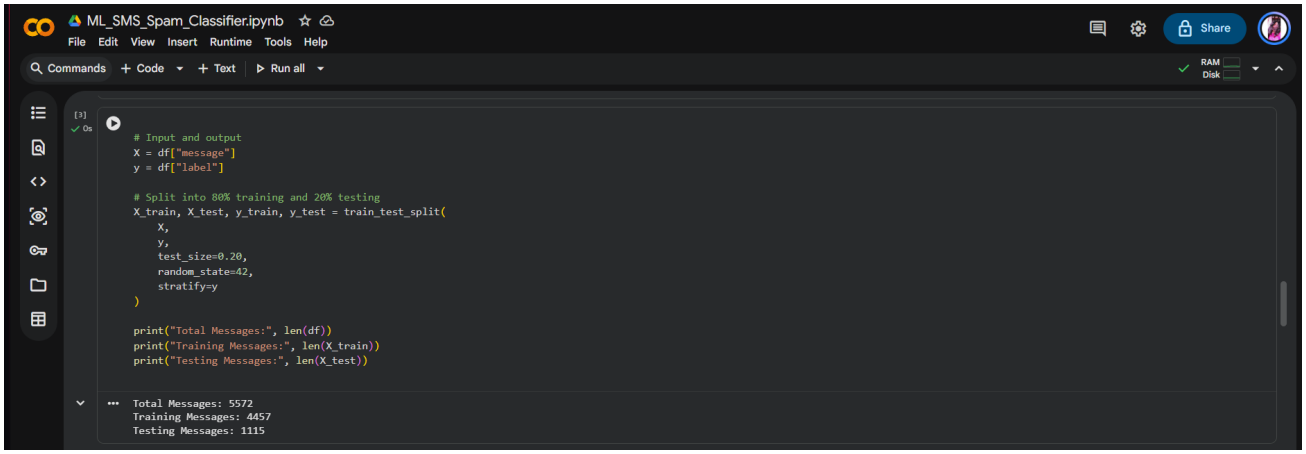


Figure 3. Output confirming the 80:20 split.

7. TF-IDF Feature Extraction

The TfidfVectorizer converts message text into numerical features. It is fitted on the training messages only, then applied to the test messages. The training matrix has shape 4457 × 7403 and the test matrix has shape 1115 × 7403, meaning 7,403 TF-IDF features were learned from the training vocabulary.

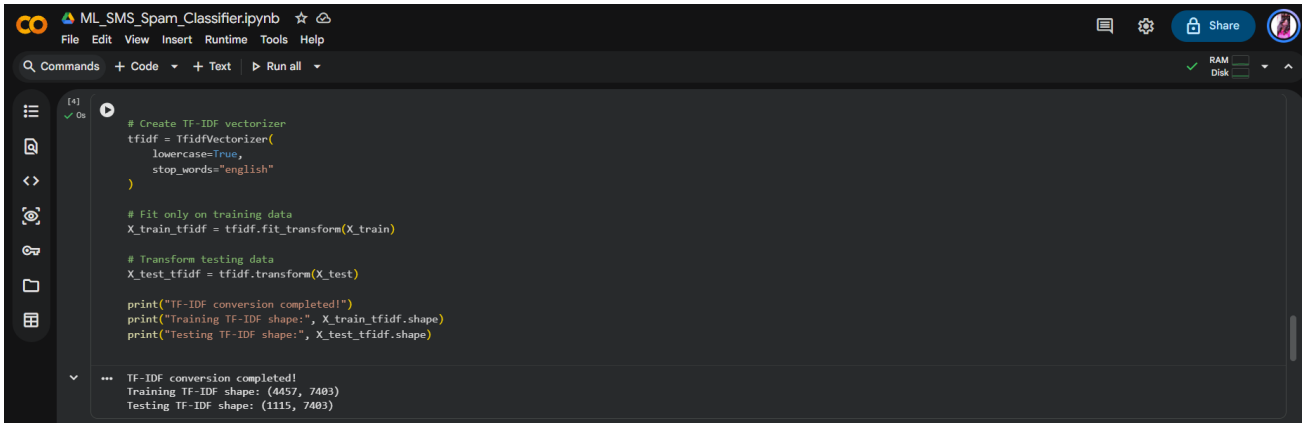


Figure 4. TF-IDF conversion and matrix dimensions.

8. Model Training and Evaluation

A Multinomial Naive Bayes classifier was trained on the TF-IDF training features. Evaluation was performed on the held-out test set of 1,115 messages.

Metric	Observed result	Interpretation
Accuracy	97.04%	Percentage of all test messages classified correctly.
Spam precision	100.00%	Predicted spam messages were correct in this test set.
Spam recall	77.85%	The model detected about 77.85% of actual spam messages.

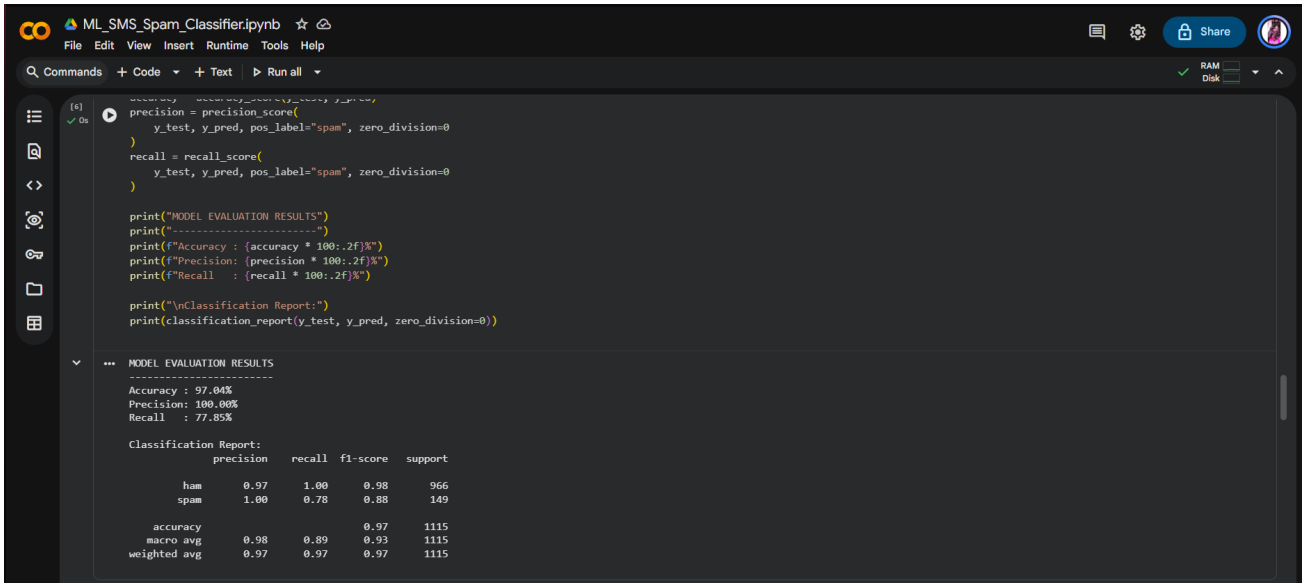


Figure 5. Classification report and evaluation metrics from the executed notebook.

Interpretation: The 100% spam precision means that, among the messages predicted as spam in the test set, none were actually ham. The 77.85% spam recall indicates that some spam messages were missed and classified as ham. Accuracy should therefore be considered together with precision and recall.

9. Confusion Matrix

The confusion matrix uses actual labels as rows and predicted labels as columns, in the order ham then spam. The values from the output are shown below.

Actual / Predicted	Predicted Ham	Predicted Spam
Actual Ham	966	0
Actual Spam	33	116

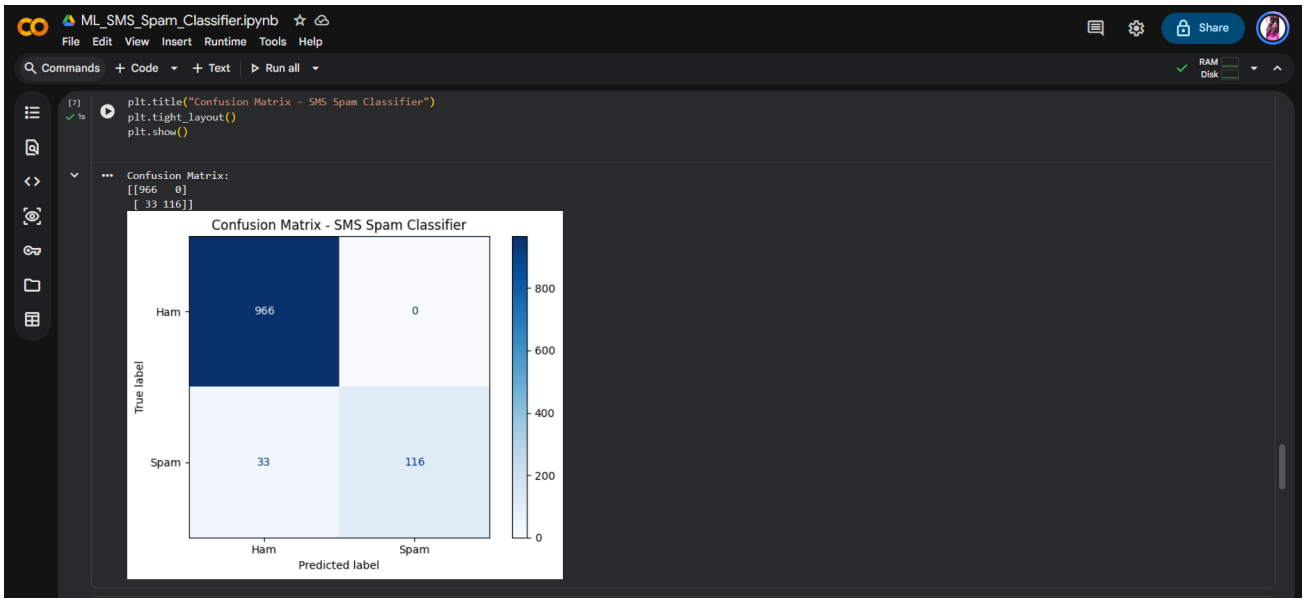


Figure 6. Confusion matrix from the test set.

The classifier correctly identified 966 ham messages and 116 spam messages. It incorrectly labelled 33 spam messages as ham, and it incorrectly labelled zero ham messages as spam. These results match the reported precision and recall values after rounding.

10. Testing on Three New Messages

Three new messages were transformed using the already-fitted TF-IDF vectorizer and classified by the trained model. The observed predictions are listed below.

Test message	Prediction
“Congratulations! You won a free prize. Claim now!”	SPAM
“Hey, are you coming to college tomorrow?”	HAM
“URGENT! You have won cash. Click the link now!”	SPAM

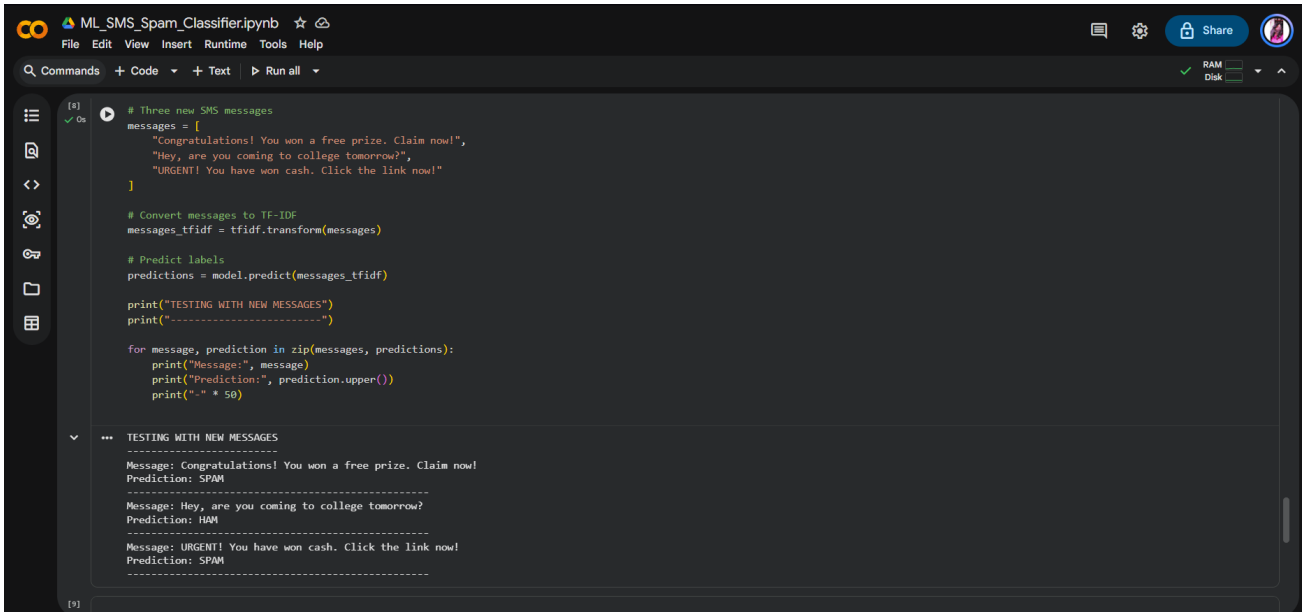


Figure 7. Predictions for three new SMS messages.

11. Conclusion

The project successfully trained a Multinomial Naive Bayes SMS classifier using TF-IDF features. On the held-out test set, it achieved 97.04% accuracy, 100.00% spam precision, and 77.85% spam recall. The confusion matrix shows that the model did not falsely flag any ham messages in this test set, although it missed 33 spam messages. The three example predictions also demonstrate how the trained model can classify new messages. Further improvements could focus on increasing spam recall while maintaining high precision.

12. Files for GitHub Submission

- `ML_SMS_Spam_Classifier.ipynb` — notebook containing the code and executed outputs.
- `sms_spam_dataset.csv` — dataset exported from the notebook.
- `spam_classifier_model.pkl` — saved Naive Bayes model, if downloaded.
- `tfidf_vectorizer.pkl` — saved TF-IDF vectorizer, if downloaded.
- `Apurwa_ML_SMS_Spam_Classifier_Report.pdf` — this report with screenshots.

Report prepared from the output screenshots supplied by the student. Student details are shown on the cover page.